

Applications of the Derivative

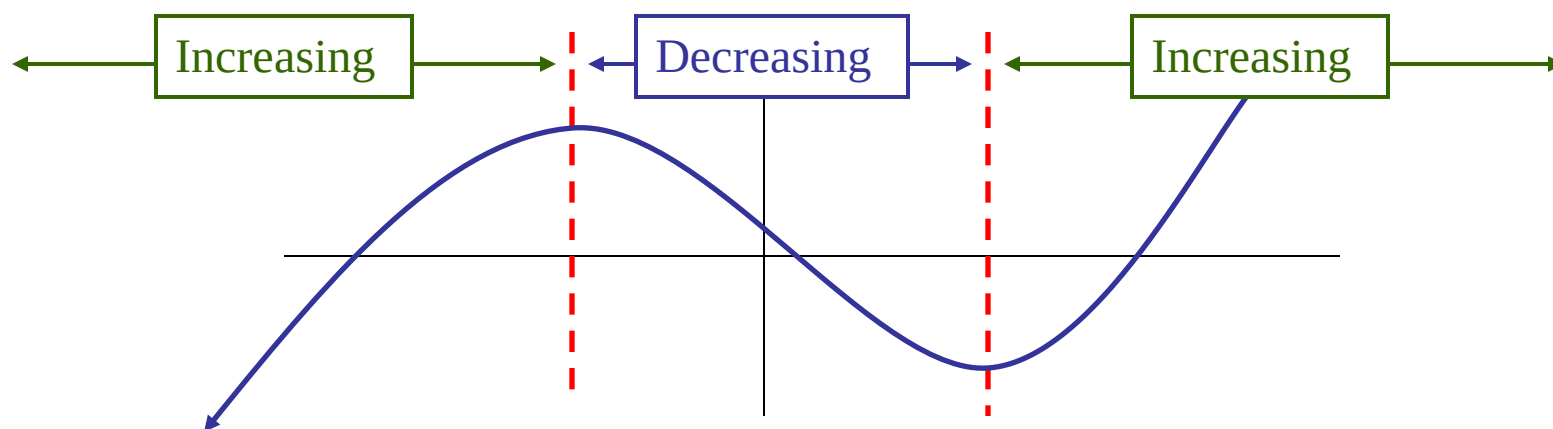
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- Applications of the First Derivative
- Applications of the Second Derivative
- Optimization I
- Optimization II

Increasing/Decreasing

A function f is **increasing** on (a, b) if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.

A function f is **decreasing** on (a, b) if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.



Increasing/Decreasing/Constant



If $f'(x) > 0$ for each value of x in an interval (a, b) , then f is increasing on (a, b) .



If $f'(x) < 0$ for each value of x in an interval (a, b) , then f is decreasing on (a, b) .



If $f'(x) = 0$ for each value of x in an interval (a, b) , then f is constant on (a, b) .

Sign Diagram to Determine where $f(x)$ is Inc./Dec.

Steps:

1. Find all values of x for which $f'(x) = 0$ or $f'(x)$ is discontinuous and identify open intervals with these points.
2. Test a point c in each interval to check the sign of $f'(c)$.
 - a. If $f'(c) > 0$, f is *increasing* on that interval.
 - b. If $f'(c) < 0$, f is *decreasing* on that interval.

Example

Determine the intervals where $f(x) = x^3 - 6x^2 + 1$ is increasing and where it is decreasing.

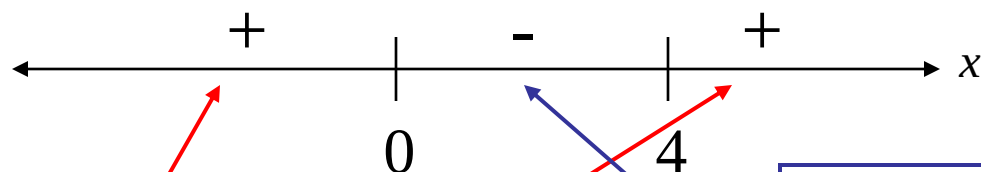
$$f'(x) = 3x^2 - 12x$$

$$3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

$$3x = 0 \text{ or } x - 4 = 0$$

$$x = 0, 4$$



f is increasing

on $(-\infty, 0) \cup (4, \infty)$

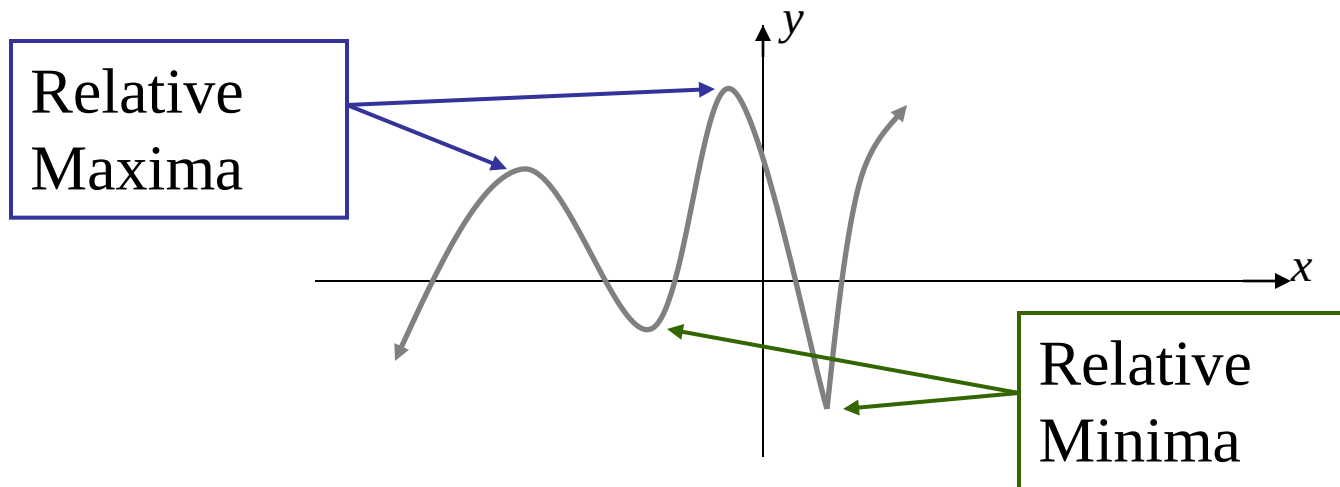
f is decreasing

on $(0, 4)$

Relative Extrema

A function f has a relative maximum at $x = c$ if there exists an open interval (a, b) containing c such that $f(x) \leq f(c)$ for all x in (a, b) .

A function f has a relative minimum at $x = c$ if there exists an open interval (a, b) containing c such that $f(x) \geq f(c)$ for all x in (a, b) .

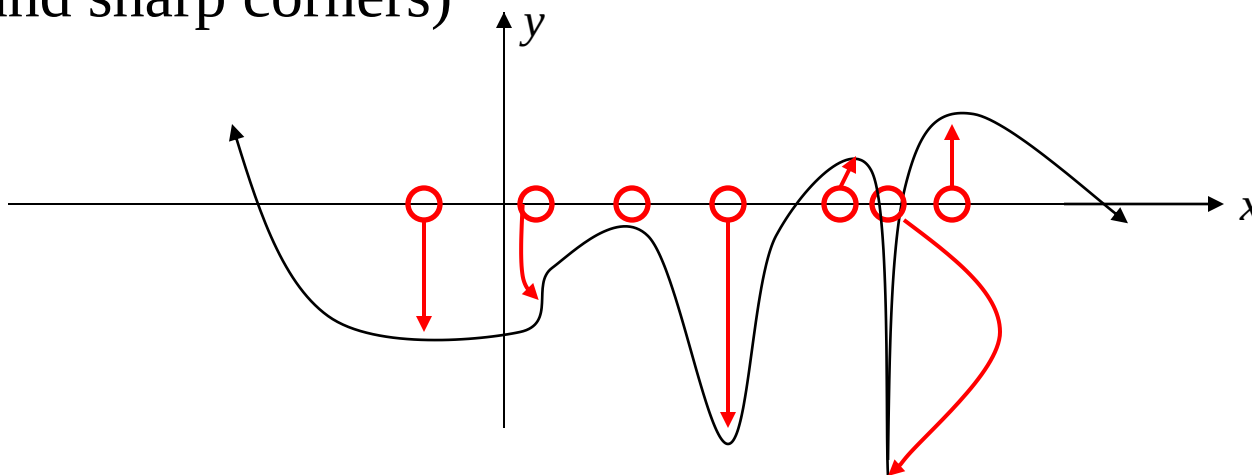


Critical Numbers of f

A **critical number** of a function f is a number in the domain of f where

$$f'(x) = 0 \text{ or } f'(x) \text{ does not exist.}$$

(horizontal tangent lines, vertical tangent lines and sharp corners)



The First Derivative Test

1. Determine the critical numbers of f .
2. Determine the sign of the derivative of f to the left and right of the critical number.

left	right	
+	—	$f(c)$ is a relative maximum
—	+	$f(c)$ is a relative minimum
No change		No relative extremum

Example

Find all the relative extrema of $f(x) = x^3 - 6x^2 + 1$.

$$f'(x) = 3x^2 - 12x$$

$$3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

$$3x = 0 \text{ or } x - 4 = 0$$

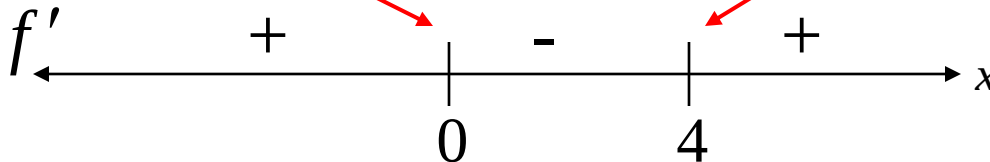
$$x = 0, 4$$

Relative max.

$$f(0) = 1$$

Relative min.

$$f(4) = -31$$



Example

Find all the relative extrema of $f(x) = \sqrt[3]{x^3 - 3x}$

$$f'(x) = \frac{x^2 - 1}{\sqrt[3]{(x^3 - 3x)^2}}$$

$$f'(x) = 0$$

$$x^2 - 1 = 0$$

or

$$x^3 - 3x = 0$$

$f'(x)$ undefined

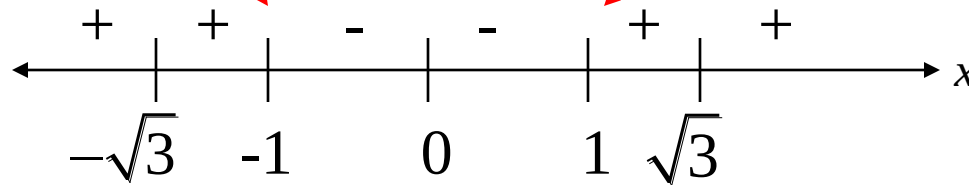
$$x = 0, \pm 1, \pm \sqrt{3}$$

Relative max.

$$f(-1) = \sqrt[3]{2}$$

Relative min.

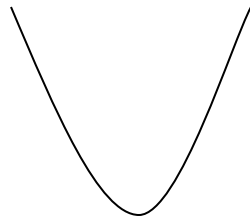
$$f(1) = -\sqrt[3]{2}$$



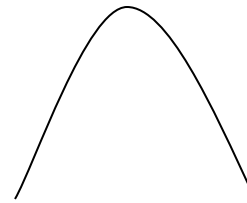
Concavity

Let f be a differentiable function on (a, b) .

1. f is *concave upward* on (a, b) if f' is increasing on (a, b) . That is, $f''(x) > 0$ for each value of x in (a, b) .
2. f is *concave downward* on (a, b) if f' is decreasing on (a, b) . That is, $f''(x) < 0$ for each value of x in (a, b) .



concave upward



concave downward

Determining the Intervals of Concavity

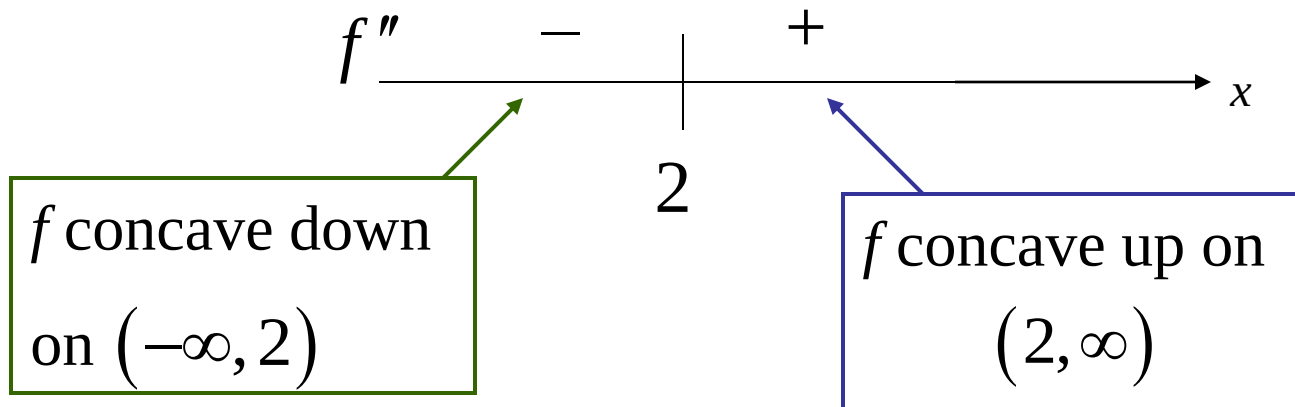
1. Determine the values for which the second derivative of f is zero or undefined. Identify the open intervals with these points.
2. Determine the sign of f'' in each interval from step 1 by testing it at a point, c , on the interval.
 $f''(c) > 0$, f is **concave up** on that interval.
 $f''(c) < 0$, f is **concave down** on that interval.

Example

Determine where the function $f(x) = x^3 - 6x^2 + 1$ is concave upward and concave downward.

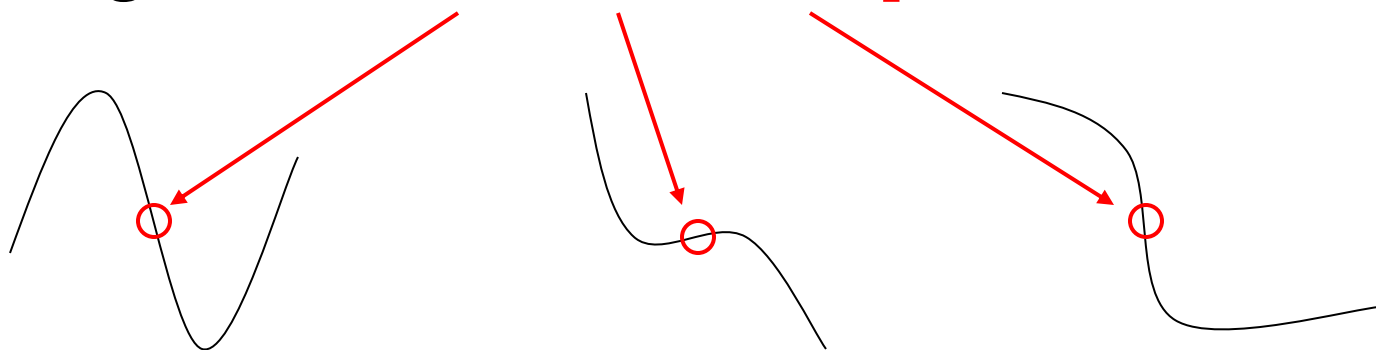
$$f'(x) = 3x^2 - 12x$$

$$f''(x) = 6x - 12 = 6(x - 2)$$



Inflection Point

A point on the graph of a continuous function f where the tangent line exists and where the concavity changes is called an inflection point.



To find inflection points, find any point, c , in the domain where $f''(x) = 0$ or $f''(x)$ is undefined.

If f'' changes sign from the left to the right of c ,
Then $(c, f(c))$ is an inflection point of f .

The Second Derivative Test

1. Compute $f'(x)$ and $f''(x)$.
2. Find all critical numbers, c , at which $f'(c) = 0$.

If	Then
$f''(c) < 0$	f has a relative maximum at c .
$f''(c) > 0$	f has a relative minimum at c .
$f''(c) = 0$	The test is inconclusive.

Example 1

Classify the relative extrema of $f(x) = x^4 - 4x^3 + 4x^2 - 5$ using the second derivative test.

$$\begin{aligned}f'(x) &= 4x^3 - 12x^2 + 8x \\ &= 4x(x-2)(x-1)\end{aligned}$$

Critical numbers: $x = 0, 1, 2$

$$f''(x) = 12x^2 - 24x + 8$$

Relative
max.

$$f(1) = -4$$

$$f''(0) = 8 > 0$$

$$f''(1) = -4 < 0$$

$$f''(2) = 8 > 0$$

Relative minima

$$f(0) = f(2) = -5$$

Vertical Asymptote

The line $x = a$ is a ***vertical asymptote*** of the graph of a function f if either

$$\lim_{x \rightarrow a^-} f(x) \text{ or } \lim_{x \rightarrow a^+} f(x)$$

is infinite.

Horizontal Asymptote

The line $y = b$ is a ***horizontal asymptote*** of the graph of a function f if

$$\lim_{x \rightarrow -\infty} f(x) = b \text{ or } \lim_{x \rightarrow \infty} f(x) = b$$

Finding Vertical Asymptotes of Rational Functions

If $f(x) = \frac{P(x)}{Q(x)}$

is a rational function, then $x = a$ is a vertical asymptote if $Q(a) = 0$ but $P(a) \neq 0$.

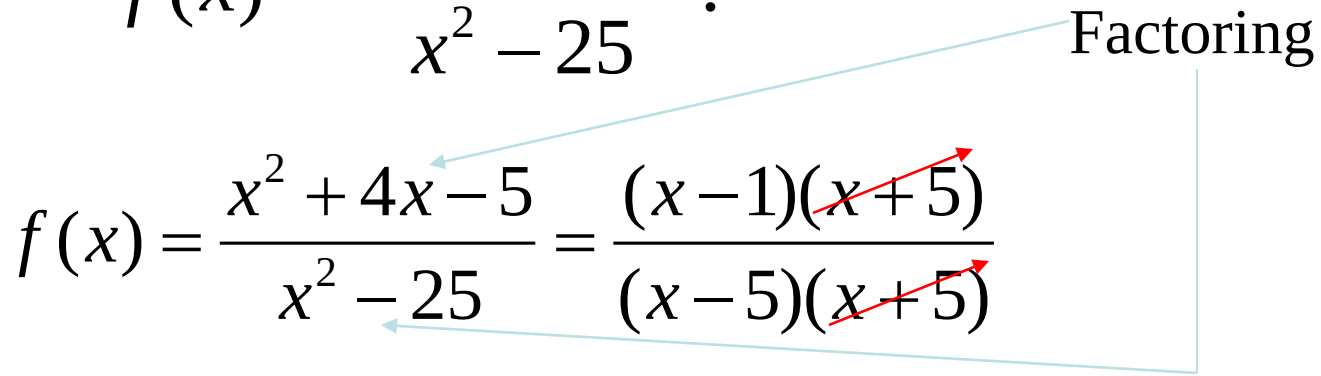
Ex. $f(x) = \frac{3x-1}{x-5}$

f has a vertical asymptote at $x = 5$.

Example

Find the vertical asymptote for the function

$$f(x) = \frac{x^2 + 4x - 5}{x^2 - 25}.$$


$$f(x) = \frac{x^2 + 4x - 5}{x^2 - 25} = \frac{(x-1)(x+5)}{(x-5)(x+5)}$$

$$f(x) = \frac{x-1}{x-5} \quad \text{when } x \neq -5$$

f has a vertical asymptote at $x = 5$.

Finding Horizontal Asymptotes of Rational Functions

Ex. $f(x) = \frac{3x^2 + 2x - 1}{x - 5x^2}$

$$\lim_{x \rightarrow \pm\infty} \frac{3x^2 + 2x - 1}{x - 5x^2} = \lim_{x \rightarrow \pm\infty} \frac{\overset{0}{3} + \overset{0}{\frac{2}{x}} - \overset{0}{\frac{1}{x^2}}}{\underset{0}{\frac{1}{x}} - 5}$$

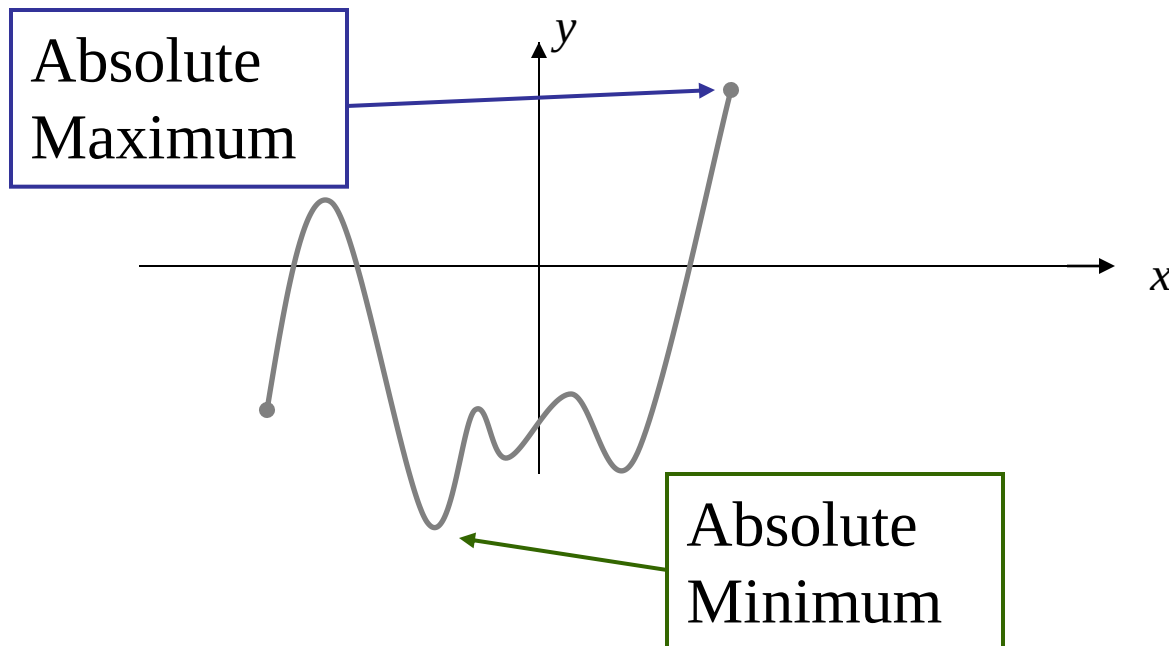
Divide
by x^2

f has a horizontal asymptote at $y = -\frac{3}{5}$.

Absolute Extrema

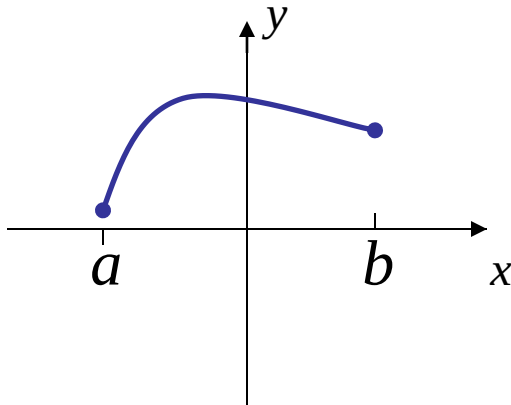
A function f has an absolute maximum at $x = c$ if $f(x) \leq f(c)$ for all x in the domain of f .

A function f has a absolute minimum at $x = c$ if $f(x) \geq f(c)$ for all x in the domain of f .

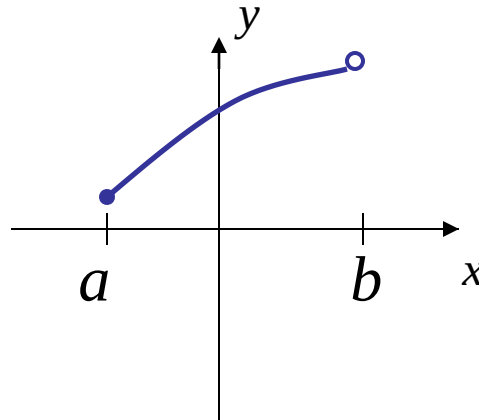


Absolute Extrema

If a function f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum and minimum on $[a, b]$.

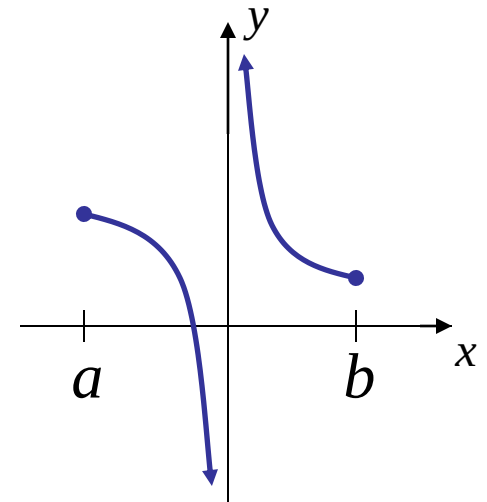


Attains max.
and min.



Attains min.
but not max.

Interval open



No min. and
no max.

Not continuous

Finding Absolute Extrema on a Closed Interval

1. Find the critical numbers of f that lie in (a, b) .
2. Compute f at each critical number as well as each endpoint.

Largest value = Absolute Maximum

Smallest value = Absolute Minimum

Example

Find the absolute extrema of $f(x) = x^3 - 3x^2$ on $\left[-\frac{1}{2}, 4\right]$.

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

Critical values at $x = 0, 2$

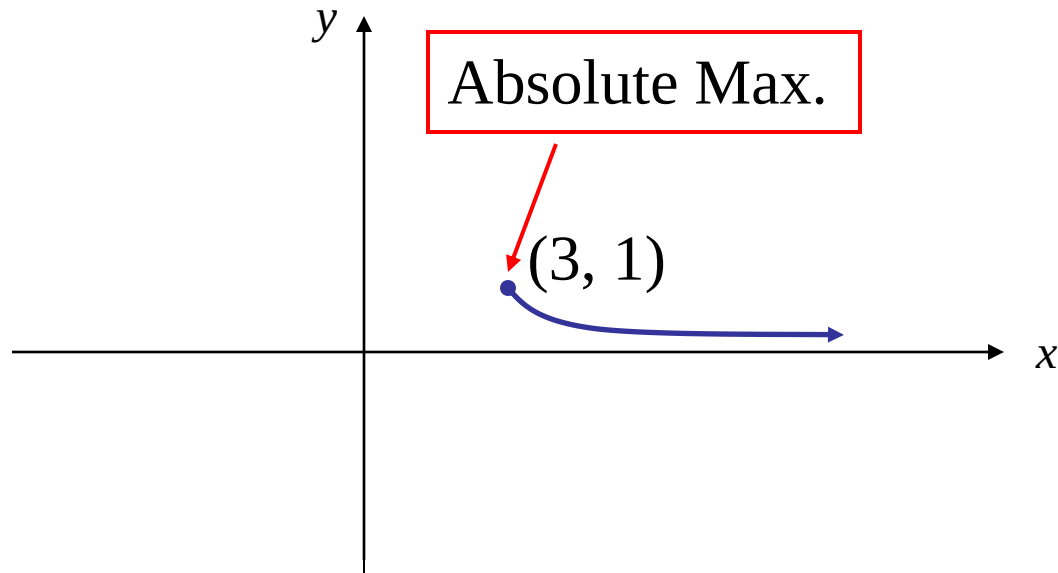
Evaluate

$$\left\{ \begin{array}{l} f(0) = 0 \\ f(2) = -4 \leftarrow \text{Absolute Min.} \\ f\left(-\frac{1}{2}\right) = -\frac{7}{8} \\ f(4) = 16 \leftarrow \text{Absolute Max.} \end{array} \right.$$

Example

Find the absolute extrema of $f(x) = \frac{1}{(x-2)}$ on $[3, \infty)$.

Notice that the interval is not closed.
Look graphically:

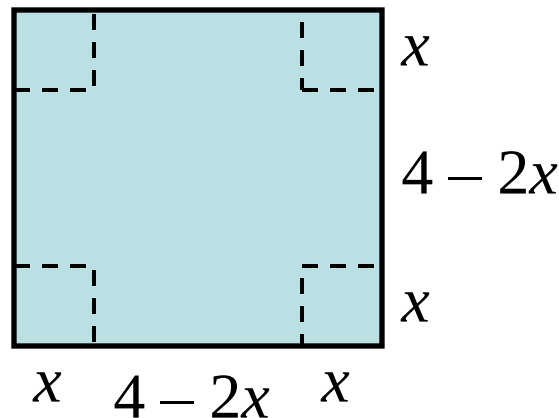


Optimization Problems

1. Assign a letter to each variable mentioned in the problem. Draw and label figure as needed.
2. Find an expression for the quantity to be optimized.
3. Use conditions to write expression as a function in one variable (note any domain restrictions).
4. Optimize the function.

Example

An open box is formed by cutting identical squares from each corner of a 4 in. by 4 in. sheet of cardboard. Find the dimensions of the box that will yield the maximum volume.



$$V = lwh = (4 - 2x)(4 - 2x)x; x \text{ in } [0, 2] \quad \dots\dots$$

$$V(x) = 16x - 16x^2 + 4x^3$$

$$V'(x) = 16 - 32x + 12x^2$$

$$= 4(2 - 3x)(2 - x)$$

Critical points: $x = 2, \frac{2}{3}$

$$V(2) = 0$$

$$V(0) = 0$$

$$V\left(\frac{2}{3}\right) \approx 4.74 \text{ in}^3$$

The dimensions are $\frac{8}{3}$ in. by $\frac{8}{3}$ in. by $\frac{2}{3}$ in.
giving a maximum box volume of 4.74 in^3 .